9.E Evaluate Limits from Expressions

Limits describe how a function behaves as the input approaches a specific value. Unlike solving equations, evaluating a limit focuses on the behavior near a point rather than the value at that point.

> *"A limit describes what a function approaches, not necessarily what it equals."*

Learning Targets

I can evaluate limits by direct substitution.
I can simplify expressions to evaluate indeterminate forms.
I can interpret limits near discontinuities.
I can explain the meaning of a limit in context.

Introduction

A limit describes the value a function approaches as x gets closer to a given number. In some cases, the function may not be defined at that point, but the limit still exists.
Evaluating limits often involves simplifying expressions and analyzing behavior near points of discontinuity.

Key Idea: A limit describes behavior near a value, not necessarily the value of the function at that point.

Connections

Prior Knowledge:
- Factoring and simplifying expressions
- Rational functions and asymptotes (9.B)
- Graphing behavior near discontinuities

Cross-Disciplinary Connections:
- Physics: motion approaching a position or value
- Engineering: systems approaching equilibrium
- Computer Science: behavior of functions as inputs grow large

Real-World Applications:
- Speed approaching zero when stopping
- Temperature approaching equilibrium
- Population growth approaching a limit

History

The concept of limits developed in the 17th century as mathematicians began studying motion and change. Isaac Newton and Gottfried Wilhelm Leibniz used limits to develop calculus.
Although early definitions were informal, later mathematicians refined the concept to make it precise and rigorous.
Today, limits form the foundation of calculus and are essential for understanding continuity, derivatives, and integrals.

Key Concepts

Evaluating Limits by Substitution

If a function is continuous at a value, the limit can be found by direct substitution.
Example:

$$\lim_{x\to 2}(x^2+3)$$

$$= 2^2+3 = 7$$

Indeterminate Forms

Sometimes direct substitution does not work. When substitution results in an undefined expression such as $\frac{0}{0}$, the limit must be evaluated by simplifying the expression.

Indeterminate Form:
An expression like $\frac{0}{0}$ that requires further simplification to evaluate the limit.

Simplifying Expressions

To evaluate limits involving indeterminate forms, simplify the expression using algebraic techniques such as factoring.
Example:

$$\lim_{x\to 2}\frac{x^2-4}{x-2}$$

Factor the numerator:

$$\frac{(x-2)(x+2)}{x-2}$$

Simplify:

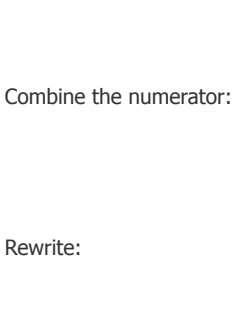
$$= x+2$$

Now evaluate:

$$\lim_{x\to 2}(x+2) = 4$$

Limits and Discontinuities

A function may be undefined at a value, but the limit can still exist if the function approaches the same value from both sides.



Key Idea: Limits describe behavior near a point, even when the function is not defined at that point.

Rationalizing to Evaluate Limits

When a limit involves a square root, rationalizing the expression can eliminate an indeterminate form.
Example:

$$\lim_{x\to 4}\frac{\sqrt{x}-2}{x-4}$$

Multiply by the conjugate:

$$\frac{\sqrt{x}-2}{x-4}\cdot\frac{\sqrt{x}+2}{\sqrt{x}+2}$$

Simplify:

$$= \frac{x-4}{(x-4)(\sqrt{x}+2)}$$

$$= \frac{1}{\sqrt{x}+2}$$

Evaluate:

$$\lim_{x\to 4}\frac{1}{\sqrt{x}+2} = \frac{1}{4}$$

Complex Fractions

Limits involving complex fractions can be simplified by combining terms or multiplying by a common denominator.
Example:

$$\lim_{x\to 1}\frac{\frac{1}{x}-1}{x-1}$$

Combine the numerator:

$$= \frac{\frac{1-x}{x}}{x-1}$$

Rewrite:

$$= \frac{1-x}{x(x-1)}$$

Simplify:

$$= -\frac{1}{x}$$

Evaluate:

$$\lim_{x\to 1} -\frac{1}{x} = -1$$

Other Indeterminate Forms

Common Indeterminate Forms:
- $\frac{0}{0}$
- $\frac{\infty}{\infty}$

These forms require algebraic simplification before evaluating the limit.

Vocabulary

Limit:
The value a function approaches as x approaches a given number.

Continuous Function:
A function where the limit equals the function value at a point.

Indeterminate Form:
An expression such as $\frac{0}{0}$ or $\frac{\infty}{\infty}$ that requires further simplification.

Discontinuity:
A point where a function is not defined or does not behave smoothly.

Conjugate:
An expression used to rationalize radicals, such as $\sqrt{x}-2$ and $\sqrt{x}+2$.

Complex Fraction:
A fraction that contains another fraction in the numerator or denominator.

Strategies

Evaluating Limits Algebraically:
1. **Start with direct substitution.**
   If the expression is defined, substitute the value directly.
2. **Identify indeterminate forms.**
   If substitution gives $\frac{0}{0}$, further work is required.
3. **Choose a simplification strategy:**
   - Factor and simplify
   - Rationalize using conjugates
   - Simplify complex fractions
4. **Simplify completely.**
   Cancel common factors and rewrite the expression.
5. **Evaluate the simplified expression.**
   Substitute again after simplification.
6. **Interpret the result.**
   Remember the limit describes behavior near the value, not necessarily the function value.

Examples

Example 1 — Direct Substitution

Evaluate the limit.

$$\lim_{x\to 2}(x^2+3)$$

Example 2 — Factoring

Evaluate the limit by simplifying.

$$\lim_{x\to 2}\frac{x^2-4}{x-2}$$

Example 3 — Rationalizing

Evaluate the limit by rationalizing.

$$\lim_{x\to 4}\frac{\sqrt{x}-2}{x-4}$$

Example 4 — Complex Fraction

Simplify and evaluate the limit.

$$\lim_{x\to 1}\frac{\frac{1}{x}-1}{x-1}$$

Example 5 — Indeterminate Form

Identify the form and evaluate the limit.

$$\lim_{x\to 3}\frac{x^2-9}{x-3}$$

Example 6 — Limit at a Discontinuity

Evaluate the limit and interpret the result.

$$\lim_{x\to 1}\frac{x^2-1}{x-1}$$

Example 7 — Interpretation of a Limit

Explain what the limit represents.

$$\lim_{x\to 0}\frac{\sin x}{x}$$

Common Mistakes

- **Stopping at $\frac{0}{0}$:**
  This is not the answer—it means further simplification is required.
- **Not factoring completely:**
  Incomplete factoring can prevent cancellation and lead to incorrect limits.
- **Forgetting to rationalize:**
  Radical expressions often require multiplying by the conjugate.
- **Errors with complex fractions:**
  Students often simplify incorrectly when fractions appear inside fractions.
- **Confusing function value with limit:**
  The limit describes behavior near a value, not necessarily the value at that point.
- **Not re-substituting after simplifying:**
  After simplification, the limit must be evaluated again.

Summary

- Limits describe how a function behaves as x approaches a value.
- Direct substitution works when the function is continuous.
- Indeterminate forms such as $frac00$ require algebraic simplification.
- Common techniques include factoring, rationalizing, and simplifying complex fractions.
- The limit may exist even when the function is undefined at that point.

Practice Set

Level 1 — Easy

1. Evaluate the limit.

$$\lim_{x\to 1}(x^2+2)$$

2. Evaluate the limit.

$$\lim_{x\to 3}(2x+5)$$

3. Evaluate the limit.

$$\lim_{x\to 9}\sqrt{x+7}$$

4. Evaluate the limit.

$$\lim_{x\to -2}(x^2+x-1)$$

5. Evaluate the limit.

$$\lim_{x\to 4}(x^2-x+3)$$

Level 2 — Medium

6. Evaluate the limit.

$$\lim_{x\to 1}\frac{x^2-1}{x-1}$$

7. Evaluate the limit.

$$\lim_{x\to -3}\frac{x^2-9}{x+3}$$

8. Evaluate the limit.

$$\lim_{x\to 5}\frac{x^2-25}{x-5}$$

9. Evaluate the limit.

$$\lim_{x\to 2}\frac{x^2-4}{x-2}$$

10. Evaluate the limit.

$$\lim_{x\to -1}\frac{x^2-1}{x+1}$$

Level 3 — Hard

11. Evaluate the limit by rationalizing.

$$\lim_{x\to 1}\frac{\sqrt{x+3}-2}{x-1}$$

12. Evaluate the limit.

$$\lim_{x\to 0}\frac{\sqrt{x+4}-2}{x}$$

13. Evaluate the limit.

$$\lim_{x\to 9}\frac{3-\sqrt{x}}{9-x}$$

14. Evaluate the limit.

$$\lim_{x\to 4}\frac{2-\sqrt{x}}{4-x}$$

15. Evaluate the limit.

$$\lim_{x\to 16}\frac{\sqrt{x}-4}{x-16}$$

Level 4 — Challenging

16. Evaluate the limit.

$$\lim_{x\to 2}\frac{\frac{1}{x}-\frac{1}{2}}{x-2}$$

17. Evaluate the limit.

$$\lim_{x\to 3}\frac{\frac{1}{x}-\frac{1}{3}}{x-3}$$

18. Evaluate the limit.

$$\lim_{x\to 1}\frac{x-1}{\frac{1}{x}-1}$$

19. Evaluate the limit.

$$\lim_{x\to 4}\frac{\frac{x}{4}-1}{x-4}$$

20. Evaluate the limit.

$$\lim_{x\to 2}\frac{\frac{2}{x}-1}{x-2}$$

Level 5 — Justification / Conceptual

21. Explain why limits can exist at points where the function is undefined.

22. Explain why simplifying expressions is necessary when evaluating limits.

23. Justify why rationalizing removes indeterminate forms involving radicals.

24. Explain how algebra reveals behavior near discontinuities.

25. Describe how limits connect algebra and graph behavior.

Answer Key

Level 1 — Easy

1. 3
2. 11
3. $\sqrt{16}=4$
4. 1
5. 15

Level 2 — Medium

6. 2
7. -6
8. 10
9. 4
10. -2

Level 3 — Hard

11. $\frac{1}{4}$
12. $\frac{1}{4}$
13. $\frac{1}{6}$
14. $\frac{1}{4}$
15. $\frac{1}{8}$

Level 4 — Challenging

16. $-\frac{1}{4}$
17. $-\frac{1}{9}$
18. -1
19. $\frac{1}{4}$
20. $-\frac{1}{2}$

Level 5 — Justification

21–25. Answers vary (conceptual)

Detailed Example Solutions

Example 1 — Direct Substitution

$$\lim_{x\to 2}(x^2+3)$$

Substitute directly:

$$= 2^2+3 = 7$$

Example 2 — Factoring

$$\lim_{x\to 2}\frac{x^2-4}{x-2}$$

Factor:

$$\frac{(x-2)(x+2)}{x-2}$$

Simplify:

$$= x+2$$

Evaluate:

$$= 4$$

Example 3 — Rationalizing

$$\lim_{x\to 4}\frac{\sqrt{x}-2}{x-4}$$

Multiply by conjugate:

$$\frac{\sqrt{x}-2}{x-4}\cdot\frac{\sqrt{x}+2}{\sqrt{x}+2}$$

Simplify:

$$= \frac{x-4}{(x-4)(\sqrt{x}+2)}$$

$$= \frac{1}{\sqrt{x}+2}$$

Evaluate:

$$= \frac{1}{4}$$

Example 4 — Complex Fraction

$$\lim_{x\to 1}\frac{\frac{1}{x}-1}{x-1}$$

Combine numerator:

$$= \frac{\frac{1-x}{x}}{x-1}$$

Rewrite:

$$= \frac{1-x}{x(x-1)}$$

Simplify:

$$= -\frac{1}{x}$$

Evaluate:

$$= -1$$

Example 5 — Indeterminate Form

$$\lim_{x\to 3}\frac{x^2-9}{x-3}$$

Factor:

$$= \frac{(x-3)(x+3)}{x-3}$$

Simplify:

$$= x+3$$

Evaluate:

$$= 6$$

Example 6 — Discontinuity

$$\lim_{x\to 1}\frac{x^2-1}{x-1}$$

Factor:

$$= \frac{(x-1)(x+1)}{x-1}$$

Simplify:

$$= x+1$$

Evaluate:

$$= 2$$

Example 7 — Interpretation

$$\lim_{x\to 0}\frac{\sin x}{x}$$

As x approaches 0, the ratio approaches 1.
Limit = 1